

Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK II

4. STRUCTURE OF THE ATOM

• Planetary model for light ions / Ex. 4.16

Calculate the speed and radial acceleration for a ground-state electron in the hydrogen atom H. Do the same for the He^+ and the Li^{++} ions.

Solution

In the planetary model, the electron follows a circular orbit due to the Coulomb force. For uniform circular motion (i.e., neglecting that the accelerated electron radiates) Newton's II law yields

$$m \frac{v^2}{r} = \frac{e^2}{4\pi\epsilon_0} \frac{1}{r^2}$$

so that the velocity is given by

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 m r_n}}.$$

Using the angular momentum quantization condition $L = mvr_n = n\hbar$, we get $r_n = n^2 a_0$ with Bohr's radius

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2} = 0.53 \times 10^{-10} \text{ m},$$

the Coulomb constant $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$ and the electron charge $|e| = 1.602 \times 10^{-19} \text{ C}$ and the electron rest mass $mc^2 = 0.511 \text{ MeV}$ one finds for $n = 1$

$$v = 7.30 \times 10^{-3} c = 2.19 \times 10^6 \text{ m/s}$$

and radial (or centripetal) acceleration is given by

$$a_c = \frac{v^2}{a_0} = 9.07 \times 10^{22} \text{ m/s}^2.$$

For other single-electron ions, the Coulomb force is modified as

$$m \frac{v^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2}$$

but also the radius changes as $r = a_0/Z$, so that the velocity $v \propto Z$ and the acceleration $a_c = v^2/r \propto Z^3$. In particular, for Li^{++} one finds $v = 2.19 \times 10^{-2} c = 6.57 \times 10^6 \text{ m/s}$ and $a_c = 2.45 \times 10^{24} \text{ m/s}^2$.

• Glue of the planetary model / Ex. 4.17

Compute and compare the electrostatic and gravitational forces in the classical hydrogen atom, assuming a radius $a_0 = 0.53 \times 10^{-10} \text{ m}$.

Solution

The Coulomb force is given by

$$F_c = \frac{1}{4\pi\epsilon_0} \frac{e^2}{a_0^2} = 8.21 \times 10^{-8} \text{ N}$$

whereas the gravitational force is given by

$$F_G = G \frac{m_e m_p}{a_0^2} = 3.61 \times 10^{-47} \text{ N},$$

calculated using for the Coulomb constant $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$, and the gravitational constant $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, and for the masses $m_e = 9.11 \times 10^{-31} \text{ kg}$ and $m_p = 1.67 \times 10^{-27} \text{ kg}$.

Therefore, the gravitational force is negligible with respect to the Coulomb force, the ratio being

$$\frac{F_c}{F_G} = 2.28 \times 10^{39}.$$

• **Angular momentum in Bohr's hydrogen atom model / Ex. 4.20**

(a) Calculate the angular momentum in kgm^2/s for the lowest electron orbit in the hydrogen atom. Compare the result with Planck's constant h . (b) Repeat for an electron in the n -th state of hydrogen.

Solution

The angular momentum is defined as

$$|\vec{L}| = |\vec{p} \times \vec{r}| = m_e v r_n.$$

(a) For $n = 1$ the velocity is given by (cfr. previous exercise) $v_0 = 7.30 \times 10^{-3} c = 2.19 \times 10^6 \text{ m/s}$, and $m_e = 9.11 \times 10^{-31} \text{ kg}$, $a_0 = 0.53 \times 10^{-10} \text{ m}$, thus yielding $L = \hbar = 1.0554 \times 10^{-34} \text{ kgm}^2/\text{s}$.

(b) For a different n , one has $v_n = v_0/n$ because $r_n = n^2 a_0$ so that

$$L = m_e v_n r_n = n \cdot (m_e v_0 a_0) = n \hbar.$$

• **Binding energy of light ions / Ex. 4.22**

What is the total mechanical energy for a ground-state electron in H, He^+ and the Li^{++} atoms? For which atom is the electron most strongly bound? Why?

Solution

The total mechanical energy of the ground-state ($n = 1$, $r_n = n^2 a_0$) is given by

$$E = K + V = \frac{1}{2} m v^2 - \frac{1}{4\pi\epsilon_0} \frac{Z e^2}{r}$$

where Z is the atomic number.

Using the expressions for the velocity and the radius of the circular orbit $r = a_0/Z$, one finds a relation between kinetic and potential energy: $V = -2K$ (which is a manifestation of the *virial theorem*) and therefore

$$E_n = -K = -\frac{1}{2} m v^2 = -\frac{Z e^2}{8\pi\epsilon_0 a_0 n^2} = -Z^2 \frac{E_0}{n^2},$$

where

$$E_0 = \frac{m e^4}{2\hbar^2 (4\pi\epsilon_0)^2} = 13.6 \text{ eV}.$$

In particular for the ground-state energies with $n = 1$, one finds

- $E_{\text{H}} = -E_0 = -13.6 \text{ eV}$
- $E_{\text{He}^+} = -4E_0 = -54.4 \text{ eV}$
- $E_{\text{Li}^{++}} = -9E_0 = -122.5 \text{ eV}$

The binding energy is larger for heavier atoms (i.e., with larger Z) due to the greater attractive force between the nucleus and electron.

• **Absorption transition / Ex. 4.23**

A hydrogen atom in an excited state absorbs a photon of wavelength $\lambda = 410\text{ nm}$. What were the initial and final states of the hydrogen atom? Use Rydberg constant in the infinite nucleus mass approximation: $R_\infty = 1.097\,373 \times 10^7\text{ m}^{-1}$.

Solution

The energy of the absorbed photon is given by

$$E_\gamma = hf = h\frac{c}{\lambda} = 3.02\text{ eV}$$

where $hc = 1240\text{ eV nm}$ was used. The transition is described by the Rydberg equation

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_\ell^2} - \frac{1}{n_u^2} \right).$$

Since the energy of $E_3 = -1.51\text{ eV}$, the transition must necessarily involve $n_\ell = 2$ ($E_2 = -3.40\text{ eV}$), then the energy difference E_γ is found for $n_u = 6$ ($E_6 = -0.38\text{ eV}$).

Alternative brute force solution: We can convert the problem into dimensionless units by dividing the Rydberg equation with the Rydberg constant:

$$\frac{1}{R_\infty \lambda} = \frac{1}{n_\ell^2} - \frac{1}{n_u^2}.$$

The left hand side is $1/(\lambda R_\infty) = 0.2222$, and to find the quantum numbers, we find the corresponding entry in the table below listing the possible values of the right hand side.

	$1/n_\ell^2 - 1/n_u^2$					
$n_\ell \backslash n_u$	1	2	3	4	5	6
1		0.75	0.8889	0.9375	0.96	0.9722
2			0.1389	0.1875	0.21	0.2222
3				0.0486	0.0711	0.0833
4					0.0225	0.0347
5						0.0122

• **Emission transition / Ex. 4.24**

A hydrogen atom in an excited state emits a photon of wavelength $\lambda = 95\text{ nm}$. What are the initial and final states of the hydrogen atom? Use Rydberg constant in the infinite nucleus mass approximation: $R_\infty = 1.097\,373 \times 10^7\text{ m}^{-1}$.

Solution

The energy of the emitted photon is given by

$$E_\gamma = hf = h\frac{c}{\lambda} = 13.05\text{ eV}$$

where $hc = 1240\text{ eV nm}$ was used. The transition is described by the Rydberg equation

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_\ell^2} - \frac{1}{n_u^2} \right).$$

Because of such high energy difference, the lowest level must necessarily involve $n_\ell = 1$ ($E_1 = -13.6\text{ eV}$), then the energy difference E_γ is found for $n_u = 5$ ($E_5 = -0.54\text{ eV}$).

• **Isotope shift of the Balmer series / Ex. 4.26**

The *isotope shift* of spectral lines refers to the shift in wavelengths (or frequencies) due to the different isotopic masses of given elements. Find the isotope shifts in wavelength for deuterium and tritium compared with hydrogen for each of the four visible lines of the Balmer series.

Solution

Within Bohr's atom model, the Rydberg equation was derived assuming the nucleus mass M to be infinite. The resulting Rydberg constant is $R_\infty = 1.097\,373 \times 10^7 \text{ m}^{-1}$. To include finite nucleus mass corrections one can recast the two-body problem an effective one-body problem, in which the electron-nucleus system revolves around its center of mass.

The *reduced mass* of the electron is given by

$$\mu_e = \frac{m_e M}{m_p + M} = \frac{m_e}{1 + \frac{m_e}{M}},$$

where M is the mass of the nucleus, and the energy levels of the electron are given by the same formulas, but with m_e replaced by μ_e (see at the end of the exercise a derivation).

Recall that the Rydberg constant is given by

$$R_\infty = \frac{E_0}{hc} = \frac{m_e e^4}{4\pi c \hbar^3 (4\pi \epsilon_0)^2},$$

but the important information in this context is that it is proportional to the electron mass m_e . Then, the corrected Rydberg constant is given by

$$R = \frac{\mu_e}{m_e} R_\infty.$$

The electron mass is $m_e = 0.0005485799 \text{ u}$ but one needs to calculate the reduced electron mass for each isotope. Recall that $1 \text{ u} = 1 \text{ Da}$ (Dalton) which is defined in terms of the mass of the Carbon-12 isotope (most abundant stable isotope) $m(^{12}\text{C})/12$, and $1 \text{ u} \approx m_p = 931.49 \text{ MeV}/c^2$ (or m_n).

In the case of hydrogen, the nucleus consists of a single proton, for deuterium it consists of a proton and a neutron, and for tritium it consists of a proton and two neutrons. Note that the total nuclear charge always balances the electric charge, and the isotopes are therefore charge neutral.

- hydrogen: $M = 1.007276 \text{ u}$ yielding $R_H = 0.99946 R_\infty$
- deuterium: $M = 2.013553 \text{ u}$ yielding $R_D = 0.99973 R_\infty$
- tritium: $M = 3.015500 \text{ u}$ yielding $R_T = 0.99982 R_\infty$

As expected, the heavier the nucleus, the more accurate the approximated value R_∞ becomes.

With the corresponding Rydberg constants, it is finally possible to calculate the isotope-shifted wavelengths. The transitions are described by the Rydberg equation

$$\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{k^2} \right), \quad k > n,$$

and the Balmer series includes four spectral lines in the visible range, corresponding to $n = 2$ and $k = \{3, 4, 5, 6\}$, whose wavelengths are given by

- k=3: $\lambda_H = 656.47 \text{ nm}$, $\lambda_D = 656.29 \text{ nm}$, $\lambda_T = 656.23 \text{ nm}$
- k=4: $\lambda_H = 486.27 \text{ nm}$, $\lambda_D = 486.14 \text{ nm}$, $\lambda_T = 486.10 \text{ nm}$
- k=5: $\lambda_H = 434.17 \text{ nm}$, $\lambda_D = 434.05 \text{ nm}$, $\lambda_T = 434.02 \text{ nm}$
- k=6: $\lambda_H = 410.29 \text{ nm}$, $\lambda_D = 410.18 \text{ nm}$, $\lambda_T = 410.15 \text{ nm}$

Finally, a historical note. While the atomic masses of different elements were measured at that time, the deuterium and tritium isotopes, as well as the neutron itself, were only discovered in the 1930s.

Optional: Let us show that one can relax the infinite-mass approximation for the nucleus in Bohr's hydrogen model, by using an effective one-body problem with a reduced mass for the electron.

Derivation from Newton's laws. From Newton's II law, the force exerted by the nucleus on the electron is

$$F_{n \rightarrow e} = m_e a_e$$

and the force exerted by the electron on the nucleus is

$$F_{e \rightarrow n} = M a_n$$

and according to Newton's III law

$$F_{e \rightarrow n} = M a_n = -F_{n \rightarrow e} = -m_e a_e.$$

One can express a_n in terms of a_e as

$$a_n = -\frac{m_e}{M} a_e.$$

Then the relative acceleration is given by

$$a_{\text{rel}} = a_e - a_n = a_e \left(1 + \frac{m_e}{M}\right) = \left(\frac{m_e + M}{m_e M}\right) m_e a_e = \frac{1}{\mu_e} F_{n \rightarrow e}.$$

Since the time derivative is a linear operation, it follows that

$$a_{\text{rel}} = a_e - a_n = \frac{d^2 r_e}{dt^2} - \frac{d^2 r_n}{dt^2} = \frac{d^2}{dt^2} (r_e - r_n) = \frac{d^2 r_{\text{rel}}}{dt^2}.$$

Comparing the two expressions for a_{rel} yields

$$\frac{d^2 r_{\text{rel}}}{dt^2} = a_{\text{rel}} = \frac{1}{\mu_e} F_{n \rightarrow e},$$

thus reducing the two-body problem to a single degree of freedom (the relative coordinate r_{rel}) and the electron moves with respect to the nucleus as a particle of mass equal to the reduced mass μ_e .

Alternative derivation. Deriving Bohr's atom model consists of two steps: (i) express the energy of the electron with its angular momentum L , and (ii) demand quantization of the angular momentum to $L = n\hbar$.

To include the reduced mass correction, first we show using classical mechanics that the relation between the *total* angular momentum and *total* mechanical energy of the two-body system (including both the electron and the nucleus) is the same as for the one-body system (fixed nucleus/infinite M), but with m_e replaced by μ_e :

$$E_{\text{tot}} = -\frac{\mu_e e^4}{2(4\pi\epsilon_0)^2 L_{\text{tot}}^2}.$$

Then, we demand quantization of the total angular momentum $L_{\text{tot}} = n\hbar$.

• **Muonic atom / Ex. 4.31**

A muonic atom consists of a muon (mass $m_\mu = 106 \text{ MeV}/c^2$ and charge $q = -|e|$) in place of an electron. For the muon in a hydrogen atom, what is (a) the smallest radius and (b) the binding energy of the muon in the ground state? (c) Calculate the series limit of the wavelength for the first three series.

Solution

(a) The smaller radius is the Bohr radius a_0 calculated with the proper reduced mass for the muon

$$\mu = \frac{m_\mu M}{m_\mu + M} = 95.2 \text{ MeV}/c^2,$$

calculated with $M = m_p = 938 \text{ MeV}/c^2$, and the corresponding Bohr radius is

$$r = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2} = 2.84 \times 10^{-13} \text{ m},$$

obtained using $\hbar = 6.58 \times 10^{-16} \text{ eV s}$, the Coulomb constant $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$, and the electron charge $|e| = 1.602 \times 10^{-19} \text{ C}$. Note that for the muonic hydrogen atom, Bohr's radius is significantly smaller than for the (traditional) electronic hydrogen because the muon is heavier than the electron, and the center of mass is shifted towards the geometrical center of the muon-nucleus system.

(b) The binding energy is given by

$$E = -\frac{e^2}{8\pi\epsilon_0 r} = -2.54 \text{ keV}$$

which is nearly two orders of magnitude larger than for traditional hydrogen.

(c) The wavelengths of the transitions for the first three series ($n = 1$: Lyman, $n = 2$: Balmer, and $n = 3$: Paschen) are obtained from Rydberg formula

$$\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{k^2} \right) = \frac{hc}{E} \left(\frac{1}{n^2} - \frac{1}{k^2} \right),$$

and for each series, the limiting transition corresponds to the ionization process of the electron (or rather the muon in this case) at $k = \infty$ so

- $n = 1$: $\lambda = hc/E = 0.49 \text{ nm}$
- $n = 2$: $\lambda = 4hc/E = 1.96 \text{ nm}$
- $n = 3$: $\lambda = 9hc/E = 4.40 \text{ nm}$

where $hc = 1240 \text{ eV nm}$ was used.

• **Positronium / Ex. 4.32**

Positronium is an atom composed of an electron and a positron (mass $m_{e^+} = m_e$, charge $q = |e|$). Calculate (a) the distance between the particles and (b) the energy of the lowest energy state of positronium. *Hint*: What is the reduced mass of the two particles?

Solution

Since the mass of electron and positron are identical, the reduced mass is

$$\mu = \frac{m_{e^+} m_e}{m_{e^+} + m_e} = \frac{m_e}{2}.$$

(a) The corresponding Bohr radius is given by

$$r = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2} = 2a_0 = 1.06 \times 10^{-10} \text{ nm}.$$

(b) The binding energy is given by

$$E = -\frac{e^2}{8\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 (2a_0)} = -\frac{|E_0|}{2} = -6.8 \text{ eV}.$$

5. WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

• Comparing de Broglie wavelength for atomic and biological matter / Ex. 5.17

Calculate the de Broglie wavelength of (a) an oxygen (O_2) molecule darting around the room at $v = 480 \text{ m/s}$ and (b) an *Escherichia coli* bacterium of mass $m_b = 6.5 \times 10^{-13} \text{ kg}$ which has been measured to move at a speed of $v_b = 1.0 \times 10^{-5} \text{ m/s}$. *Optional:* Compare the de Broglie wavelength and the object's size in each case.

Solution

(a) The most abundant stable isotope of oxygen, ^{16}O has a mass

$$m_{\text{O}} = 15.995 \text{ u} \approx 16 \text{ u} = 26.576 \times 10^{-27} \text{ kg}$$

and hence, one can estimate the mass of the O_2 molecule as

$$m_{\text{O}_2} \approx 32 \text{ u} = 53.152 \times 10^{-27} \text{ kg}.$$

Recall that $\text{u} = 1.661 \times 10^{-27} \text{ kg}$ is the atomic mass unit, defined as $1/12$ of the mass of ^{12}C , and $m_{\text{p}} = 1.00728 \text{ u}$. Then the de Broglie wavelength is given by

$$\lambda = \frac{h}{p} = 2.60 \times 10^{-11} \text{ m} = 0.26 \text{ \AA}.$$

The *kinetic diameter* (i.e., a measure of the size of the molecule related to likelihood of colliding with another molecule in a gas) is $d = 2.96 \text{ \AA}$, which is comparable with its de Broglie wavelength.

(b) The *Escherichia coli* is one of the bacteria that is found in the lower intestine of warm-blooded organisms, it is harmless and even beneficial to humans. Cells are typically rod-shaped, and are about $\ell = 2 \mu\text{m}$ long and $d = 1 \mu\text{m}$ in diameter.

In this case, the mass of the bacteria is given explicitly, so it is trivial to calculate the de Broglie wavelength as

$$\lambda = \frac{h}{p} = 1.02 \times 10^{-16} \text{ m} = 0.102 \text{ fm}.$$

Since $\lambda \ll d$, quantum effects are not observable for these bacteria.

In both cases, the value $h = 6.626 \times 10^{-34} \text{ J s}$ for Planck's constant was used.

• De Broglie wavelength of a rock / Ex. 5.5

Calculate the de Broglie wavelength of a rock with mass $m = 1.2 \text{ kg}$ thrown with a speed $v = 6.0 \text{ m/s}$ into a pond. Is this wavelength similar to that of the water waves produced? Explain.

Solution

The de Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv} = 9.2 \times 10^{-35} \text{ m},$$

where $h = 6.626 \times 10^{-34} \text{ J s}$ is Planck's constant.

No, this wavelength is much-much smaller; also the wavelength of the water waves depends on the medium, as they are strictly mechanical waves.

• Spatial resolution of TEM / Ex. 5.7

Transmission electron microscopes that use high-energy electrons accelerated over a range of voltages from $\Delta V = 40 \text{ kV}$ to $\Delta V = 100 \text{ kV}$ are employed in many applications including the study of biological samples (for example, a virus) and nanoscience research and development (alloy particles and carbon nanotubes, for example). What would be the spatial limitation for this range of electrons? It is often true that resolution is limited by the optics of the lens system, not by the intrinsic limitation due to the de Broglie wavelength.

Solution

The kinetic energy of the accelerated electrons is $K = e\Delta V$. Assuming relativistic effects are small and we can use the classical energy-momentum relation,

$$\lambda_{\text{classical}} = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{hc}{\sqrt{2(mc^2)K}}.$$

For $K = 40 \text{ keV}$ one obtains $\lambda_{\text{classical}} = 6.13 \text{ pm}$. For $K = 100 \text{ keV}$ one obtains $\lambda_{\text{classical}} = 3.87 \text{ pm}$. This is a very small length, the practical resolution of TEM's is much worse than this for other technical reasons, in the range of 100 pm .

• **Phase and group velocity / Ex. 5.24**

Consider electrons of kinetic energy $K = 6.0 \text{ eV}$. Find the de Broglie wavelength λ , particle speed v , phase velocity v_{ph} , and group velocity u_{gr} .

Solution

The de Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{m_e v} = \frac{h}{\sqrt{2m_e K}} = \frac{hc}{\sqrt{2(m_e c^2)K}} = 0.501 \text{ nm}$$

where $hc = 1240 \text{ eV nm} = 1240 \times 10^{-9} \text{ eV m}$ and $m_e c^2 = 511 \text{ keV}$.

The particle velocity is given by

$$v = \frac{p}{m_e} = \frac{h}{\lambda m_e} = \frac{c}{\lambda} \frac{hc}{m_e c^2} = 1.45 \times 10^6 \text{ m/s} \quad \text{or} \quad \beta = \frac{v}{c} = 4.83 \times 10^{-3}$$

thus confirming that the non-relativistic treatment is appropriate. Note that an alternative way to compute this is $v = \sqrt{2m_e K}$.

The phase velocity is found as

$$v_{\text{ph}} = \frac{\omega}{k} = \frac{E/\hbar}{p/\hbar} = \frac{E}{p} = \frac{\frac{1}{2}m_e v^2}{m_e v} = \frac{v}{2} = 7.25 \times 10^5 \text{ m/s}.$$

The group velocity is found as

$$u_{\text{g}} = \frac{d\omega}{dk} = \frac{dE}{dp} = \frac{d}{dp} \left(\frac{p^2}{2m_e} \right) = \frac{p}{m_e} = v = 1.45 \times 10^6 \text{ m/s},$$

i.e., the group velocity is the same as the particle velocity.